RV College of Engineering®
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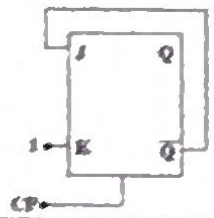
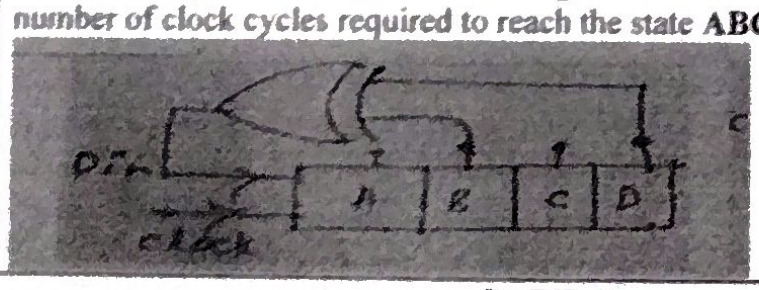
Academic year 2025-2026 (Odd Sem)
DEPARTMENT OF

# COMPUTER SCIENCE & ENGINEERING

| | | | |
|---|---|---|---|
| Date | 16-12-2025 | Maximum Marks | 60 |
| Course Code | CS234AI | Duration | 120 Min |
| Sem | III Semester | Test-II | |
| **Applied Digital Logic Design & Computer Organization (ADLD & CO)** | | | |

## Part A

| Sl. No. | Questions | M | BT | CO |
|---|---|---|---|---|
| 1.1 | How many flip-flops are required to design a Mod-20 counter? | 1 | 1 | 1 |
| 1.2 | Given the present state = 0 and next state = 1 for a JK FF, find the required values of J and K inputs. | 1 | 3 | 3 |
| 1.3 | A JK flip-flop is configured as shown in the figure. The flip-flop is initially reset (Q = 0) and is then clocked for six consecutive pulses. Determine the output sequence at Q for the six clock pulses.   | 2 | 2 | 2 |
| 1.4 | A flip-flops has a 3 ns delay. What is the maximum delay in a 10-bit binary ripple counter that uses these flip-flops? What is the maximum frequency at which the counter can operate? | 2 | 3 | 3 |
| 1.5 | A binary counter has 4 outputs, Q3, Q2, Q1, Q0, with Q3 as MSB. A NAND gate output is connected to active low clear inputs of all Flip-flops. What is the last count one can observe if Q3 and Q2 are connected as inputs to the NAND gate? | 2 | 1 | 4 |
| 1.6 | A 4 bit shift register circuit configured for right shift operation is shown below, if the present state of the shift register is **ABCD = 1101**, then the number of clock cycles required to reach the state **ABCD = 1111** is ----------   | 2 | 2 | 2 |
| | **PART- B** | | | |
| 2. a) | Describe the working principle of a JK master–slave flip-flop with suitable waveforms. Explain why the JK master–slave flip-flop eliminates the race-around condition. | 5 | 3 | 2 |
| 2. b) | Derive characteristic equation for JK and D Flip-flop and show how the D flip flop can be converted to a JK flip flop. | 5 | 3 | 2 |
| 3. a) | Construct the logic diagram of a 4-bit register with four *D flip-flops* and 4 × 1 multiplexers with mode selection input m1 and m0. The register operates according to the following function table: | 5 | 3 | 2 |

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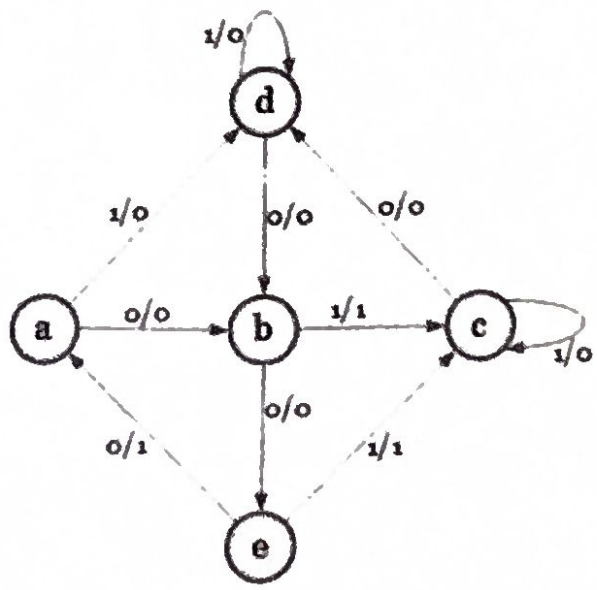
| | | | | |

| m1 | m0 | Register operation |
|---|---|---|
| 0 | 0 | No Change |
| 0 | 1 | Complement the four Output |
| 1 | 0 | Clear register to 0 (Synch) |
| 1 | 1 | Load parallel data |

| | | | | |
|---|---|---|---|---|
| 3. b) | Discuss the concept of propagation delay in ripple counters and interpret how this delay affects the maximum operating frequency. Design Negative (↓) Edge triggered MOD 6 Asynchronous up counter using T flip flop. | 5 | 2 | 3 |
| 4. | A sequential circuit with two D flip-flops A and B, two inputs, x and y ; and one output z is specified by the following next-state and output equations:<br>$A(t+1) = xy' + xB$<br>$B(t+1) = xA + xB'$<br>$z = A$<br>(a) Draw the logic diagram of the circuit.<br>(b) List the state table for the sequential circuit.<br>(c) Is it Mealy or Moore Machine? Draw the corresponding state diagram. | 10 | 4 | 4 |
| 5. a) | Determine the reduced state diagram for the given state diagram. | 5 | 3 | 1 |

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|---|---|---|---|---|
| 5. b) | Differentiate between<br>i. Mealy and Moore finite state machines.<br>ii. Ring Counter and Johnson counter | 5 | 3 | 4 |
| 6. | Design of a synchronous counter for the following count sequence: 0, 3, 5, 6 using T flip flop (Avoid Lock out condition by forcing unused states to next counting state). | 10 | 4 | 3 |

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

| Marks Distribution | Particulars | | CO1 | CO2 | CO3 | CO4 | L1 | L2 | L3 | L4 | L5 | L6 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Test | Max Marks | 06 | 19 | 18 | 17 | 03 | 09 | 28 | 20 | - | - |

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R V College of Engineering

Department of Computer Science and Engineering (data science and cyber security)

ADLDCO Retest - I : Question Paper

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|---|---|---|---|
| Course (Code)CS234 AI | Applied digital logic design and computer organisation | | Semester : 3rd BE |
| Date : Jan 2026 | Duration : 120 minutes | Staff :KBR,DD/KB PSB MH Sujatha Bedinger | |
| Name : | USN : | Section : | A B C D E CD CY |

**PART A**

| Sl. No | Quiz questions | Marks | * L1-L6 | *CO |
|---|---|---|---|---|
| 1.1 | Solve the following using minimal number of NAND gates? (A+B)(C+D) | 1 | L3 | CO2 |
| 1.2 | Design a full adder using two half adders? | 1 | L2 | CO2 |
| 1.3 | i) Subtract the following numbers using 2's complement. Assume 8 bits<br>20 from 13<br><br>ii) Construct the Booth recording of a multiplier for the following sequence.<br><br>0 1 0 1 0 1 1 0 0 1 1 0 0 0 1 1 1 1 1 1 1 0 | 1+1 =2 | L2 | CO2 |
| 1.4 | Design a 32 to 1 multiplexer using lower order multiplexers. | 2 | L2 | CO2 |
| 1.5 | Design a 3 to 8 line decoder using two, 2 to 4 line decoder and explain it. | 2 | L2 | CO1 |
| 1.6 | Represent the number +15.25 using 32 bit(single precision) floating point representation. | 2 | L2 | CO1 |

**PART B**

| Sl. No | Answer all the questions | Marks | * L1-L6 | *CO |
|---|---|---|---|---|
| 1a | A combinational switching network with inputs A, B, C, D should output 0 if the input combination is a valid Excess-3 coded decimal digit and 1 otherwise. Implement the network using basic gates after simplification using Kmap. | 4 | L2 | CO2 |
| b | Design a four bit carry look ahead adder. How does propagation delay in a Carry Look ahead Adder compare to that of parallel adder? | 6 | L2 | CO2 |
| 2a | Realize the following function using<br>i) 8 to 1 mux ii) 4 to 1 mux<br>f(a,b,c,d)= Σm(0,1,2,4,5,8,9,12,13,15) | 4 | L3 | CO2 |